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Weather and Climate

Let's Learn About

- Atmosphere
- Climate
- Elements of Weather and Climate
- Factors Influencing Climate
- Climatic Zones of the World
- Weather

Warm Up

Compare the climate of your city with another city of your choice. How similar or how different is it? Discuss in class.

Atmosphere

People live in different places in the world and follow varied lifestyles. The food they eat, the clothes they wear and the houses they live in vary from place to place. This variation is due to the condition of the weather of the place where they live.

The Earth is surrounded by a thin layer of air called the atmosphere. Just as there are layers inside the Earth, there are also layers in the atmosphere. These layers heat up or cool down, and interact with the top layer of the Earth's crust. This causes changes in the temperature of a place. That is why we sometimes feel cold and sometimes hot.

Weather

Weather is the condition or the state of the atmosphere over a short period of time at a particular place. It keeps changing from time to time. In the morning, when the sun is at the horizon, the weather is pleasant. At noon, when the sun shines brightly above our head, the weather is very hot. And again in the evening when the sun sets, the weather becomes pleasant. Weather of a place may change each day.

The weather at one place may be different from another place at the same time. For example, during the months of April, May and June, it is very hot in the Northern Hemisphere but it is cold in the Southern Hemisphere. In the months of December, January and February, it is just the opposite.



Sunny



Stormy



Few clouds



Rainy



Cloudy



Snowy

Changes in the weather

Changes in weather conditions give rise to seasons. They affect our daily activities. They also affect the clothes we wear and the food we eat.

Do and Learn

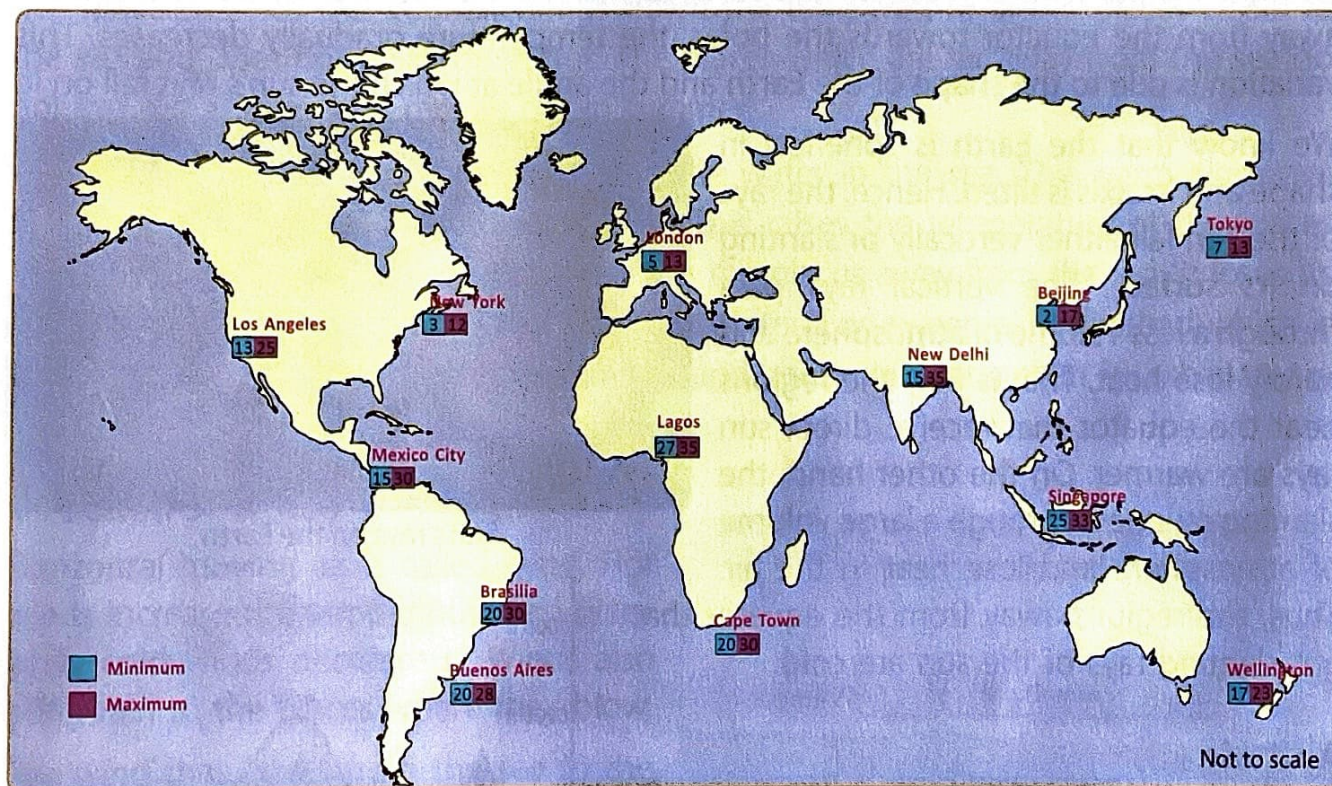
What is the study of weather known as? Find out.

Climate

Climate is the average weather condition of a place for a long period of time. It is different at different places of the world. It may rain throughout the year in some areas of the world while some areas may be dry almost the whole year round. The climate of a place affects the food habits, clothing, transportation and houses of the place. People who study climate are called climatologists.

Elements of Weather and Climate

Temperature is the measurement of warmth or coldness of air around us. The temperature and the amount of rainfall at different places of the world are not the same. As we travel far and wide, we spot differences in the climatic conditions of various places. There are many factors that affect the climate of a place.



Weather forecast showing temperature at different places on a particular day

Learn More

- Temperature is the degree of warmth or coldness of air in a region and is measured by an instrument called thermometer.
- The temperature goes down by 1°C at every 165 metres above the sea level. For example, if a place located at the sea level has the temperature of 30°C , then a place that is 165 metres above it will have the temperature of 29°C .

Humidity and Rainfall

The amount of water vapour or moisture present in the air is known as humidity. The degree of humidity of a place is also related to its distance from the sea. The closer the place is to the sea, the more humid it is. Humidity of a place increases in the rainy season, as the air carries more moisture in it then. That is why we sweat a lot in the rainy season. Humidity also affects the climate of a place as it causes rainfall. In very cold places, humidity causes snowfall instead of rainfall.

Learn More

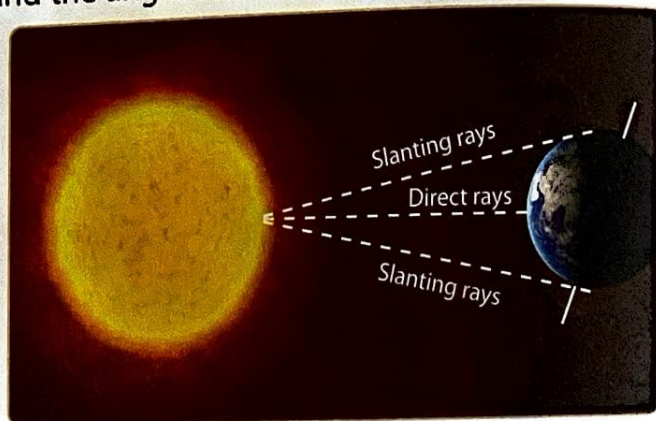
- Humidity is measured by an instrument called hygrometer.
- Rainfall, snowfall and drizzle are known as precipitation.

Factors Influencing Climate

Latitude

Latitude refers to the distance of a place from the equator. Equator receives more sunlight than any other place on the Earth as it is the closest to the sun. As we move away from the equator towards the poles, the temperature gradually decreases. This variation is due to the shape of the Earth and the angle at which the sun's rays fall on it.

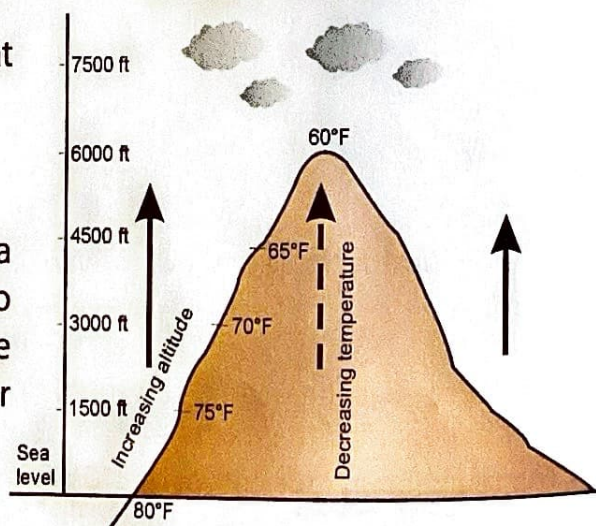
We know that the Earth is spherical in shape and its axis is tilted. Hence, the rays of the sun fall either vertically or slanting on its surface. The vertical rays pass through a less volume of atmosphere and hardly lose heat. That is why the regions near the equator that receive direct sun rays are warmer. On the other hand, the slanting rays pass through a large volume of atmosphere and lose heat in the air. Thus, the regions away from the equator that get slanting rays of the sun are cold.



Sun's rays on the Earth

Altitude

Altitude is the height of a place above the sea level. As all the seas of the world are linked to each other, their level is the same all over the Earth. Sea level is hence taken as the parameter for measuring the altitude of a place.



Altitude and temperature

The climate of a place is also affected by its altitude. The air closer to the Earth's surface is dense and absorbs more heat. This makes places at lower altitudes hot. The atmosphere is thinner at high altitudes. It absorbs less heat making the places in hilly areas cold. That is why most of the mountain peaks are covered with ice in winter and hill stations are cold even in summer. For example, Shimla which is at a high altitude is always cooler than Delhi which is at a low altitude.

Class Discussion

Why is Shimla cooler than Ludhiana, although both are at equal distance from the equator? Discuss in class.

Distance from the Sea

The distance from the sea also determines the climate of a place. Places near the sea have moderate climate but places far away from the sea are very cold in winter and very hot in summer. This is because land gets heated faster and loses heat faster than water.

During summer, land gets heated quickly and the air above the land surface becomes hot. This hot air rises up. The cool air above the sea blows towards the land to occupy the space created. This cool air brings down the temperature of the places closer to the sea and does not make them hot. The places which are far away from the sea do not get this cool air and remain hot in summer.

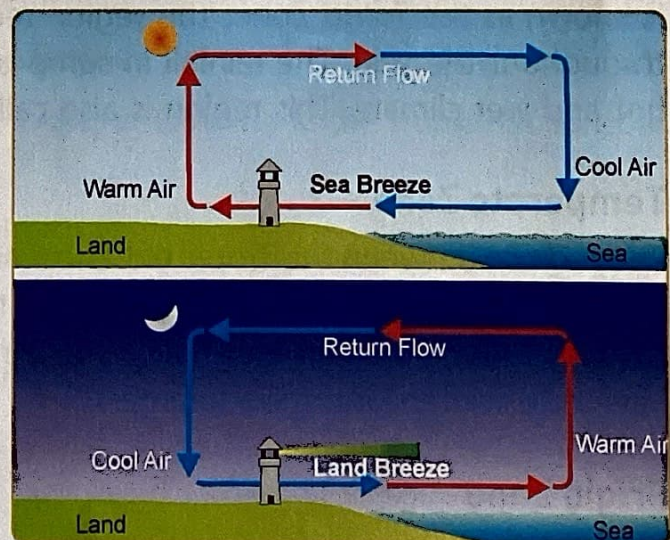
During winter, land becomes cooler than the water in the sea. The warm air above the sea rises up and moves towards land. This raises the temperature of the places along the coast. By the time this air reaches the places away from the sea, it loses its warmth. Hence, these places remain cold in winter. For example, Hyderabad which is far away from the sea is hotter in summer and cooler in winter than Visakhapatnam which is on the coast.

Direction of Winds

Horizontal moving air is called wind. Hot winds increase the temperature of a place while cold winds decrease it. Winds also bring rain to the places where they blow.

The wind that blows from the sea to the land is known as **onshore wind** or **sea breeze**. It carries moisture and makes the climate in the coastal areas humid. It also brings rain to these areas.

The wind that blows from the land to the sea is called **offshore wind** or **land breeze**. It is dry and does not carry moisture.



Land and Sea Breeze

India is affected by seasonal winds known as **monsoons**. The south-west monsoon winds bring rain from June to September and the north-east monsoon winds bring rain from October to December.

Climatic Zones of the World

The temperature and the amount of rainfall determine the climate of a place. The whole world is divided into **climatic regions** or **heat zones** according to the type of climate they have. The three main climatic zones are **Torrid**, **Temperate** and **Frigid** zones.



Climatic zones of the world

Torrid Zone

The region between the two tropics, the Tropic of Cancer and the Tropic of Capricorn, is known as the **Torrid Zone**. This region receives direct sun rays making it hot almost throughout the year. The rainfall in some places here is very heavy. Thus they have a hot and wet climate. This region is also called the **Tropical Zone**.

Temperate Zone

The regions between the Tropic of Cancer and the Arctic Circle and between the Tropic of Capricorn and the Antarctic Circle form the **Temperate Zone**. These regions receive slanting rays of the sun and hence the climate is neither too hot nor too cold. They have cold winters and mild summers.

Frigid Zone

The regions around the poles are known as **Frigid Zones**. They lie between the Arctic Circle and the North Pole and between the Antarctic Circle and the South Pole. They

are very far away from the equator and receive absolutely slanting rays of the sun. During most months in a year, the Frigid Zones do not receive the sun's rays at all. That is why they are covered with thick sheets of ice most of the year. The climate here is extremely cold throughout the year.

Let's Recall

- Weather refers to the condition or the state of atmosphere over a short period of time at a particular place.
- Climate is the average weather condition of a place over a long period of time.
- Temperature is the measurement of warmth or coldness of air around us.
- The world has been divided into three heat zones – torrid, temperate and frigid – on the basis of the temperature variations in them.
- The regions between the Tropic of Cancer and the Arctic Circle, and between the Tropic of Capricorn and the Antarctic Circle form the Temperate Zone.
- The region between the Tropic of Cancer and Tropic of Capricorn is known as Torrid Zone.
- The regions around the poles are known as Frigid Zones.

Exercises

A Tick the correct answer.

1. Who among the following study climate?

☐

Seismologists

☐

Climatologists

☐

Cartographers

2. What is the height of a place above sea level known as?

☐

Latitude

☐

Altitude

☐

Precipitation

3. Which of the following is the measurement of warmth or coldness of air around us?

☐

Temperature

☐

Precipitation

☐

Humidity

4. What is the heat zone between the Tropic of Cancer and the Tropic of Capricorn known as?

☐

Torrid

☐

Temperate

☐

Frigid

B Fill in the blanks.

1. The thin layer of the air surrounding the Earth is called the _____.
2. The amount of water vapour present in air is called _____.
3. Temperature _____ with increase in altitude.
4. The coldest climatic zone on the Earth is _____.

5

Natural Resources

Let's Learn About

- Renewable Resources
- Non-renewable Resources
- Uses of Natural Resources
- Conservation of Natural Resources

Warm Up

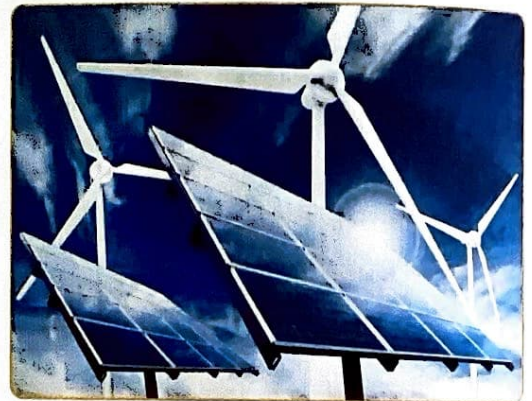
How are coal, sunlight and trees helpful to us? Discuss in class.

Natural resources are the naturally occurring substances that are considered very valuable for the survival of all living things on the Earth. Our planet is gifted with a wide range of natural resources such as water, land, forests, animals, sunlight and minerals. These resources are the gifts of nature. They are the sole reason why life is possible on our planet. The life of people over the years has shaped up according to the way in which they have used the natural resources. A nation's wealth is largely dependent on how well they use their natural resources.

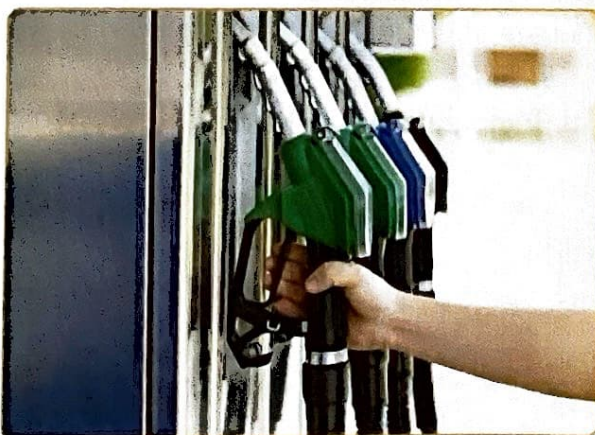
Natural resources can be of two types. They are renewable and non-renewable natural resources.

Renewable Resources

Renewable resources are the resources that can be used again and again. They are inexhaustible. They are part of our natural environment and form our ecosystem. Sunlight, water and air are some examples of renewable resources.



Renewable resources



Non-renewable resources

Non-renewable Resources

Non-renewable resources are the resources that are exhaustible and cannot be replaced. They exist only in a fixed amount and cannot be re-made. Some examples of non-renewable resources are fossil fuels, iron, uranium, petroleum and coal.

Uses of Natural Resources

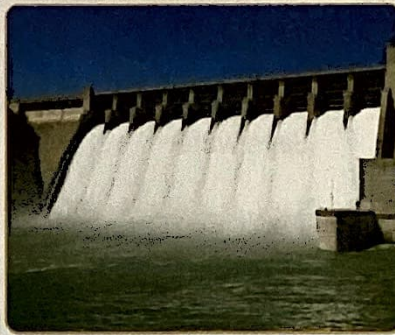
The uses of natural resources are immense. Some natural resources can be used directly and some need to be processed before use. Air, water and sunlight can be used directly. Water is used for different purposes. It is used to produce electricity by building dams. It is also used for irrigation and farming. Solar energy is another source we use directly. It is a never-ending resource. Scientists have been researching on how it can be used for different applications. Many vehicles and gadgets have been made which work with solar energy. Wind energy is also used for producing electricity.

Learn More

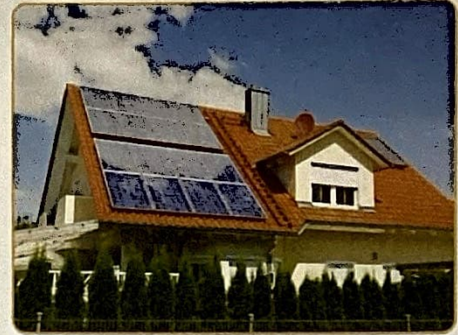
The largest oil-producing country in the world is Saudi Arabia.



Wind energy



Hydro energy



Solar energy

Some resources can't be used in their natural form. They need to be processed before they can be used. For example, petroleum that is taken out from the Earth is in a crude form. It is refined and changed to petrol, diesel, kerosene and other products. The same case is with minerals also. They need to be extracted from their ores and refined before they can be used.

All the industrial growth and development that we see around us would not have been possible without the use of natural resources.



Petroleum

Do and Learn

Find out from where do we get the electricity that we use at home or school.

Learn More

Mumbai High is India's largest offshore oilfield.

Conservation of Natural Resources

We have learned about the importance of natural resources. Since many natural resources are non-renewable, we must do something to preserve them. If we are going to use these resources continuously, they will get exhausted soon. So there is a rising

demand to conserve all our natural resources. Air, water and land are being affected due to their overuse. Many plants and animal species have become extinct and many others are on the verge of extinction. No matter how much we have progressed, we can't survive without the natural resources. Thus, we must look after the gifts that nature has given us with utmost care.



Save plants



Save water

Class Discussion

In what ways can we conserve resources in our daily lives in school as well as at home? Discuss in class.

Let's Recall

- Natural resources are the naturally occurring substances which are considered very valuable for the survival of all living things on the Earth.
- Renewable resources are resources which can be used again and again.
- Non-renewable resources are those which are exhaustible and can't be replaced.
- Some resources can be used directly and some need to be processed before use.
- We must conserve our natural resources.

Exercises

A Tick the correct answer.

- Which of the following resources is used in producing electricity by making dams?
☐ Wind ☐ Water ☐ Sunlight
- Which of the following is not a gift of nature?
☐ Minerals ☐ Water ☐ Highways
- Which of the following resources is non-renewable?
☐ Coal ☐ Sunlight ☐ Air
- Which of the following resources cannot be used directly?
☐ Petroleum ☐ Air ☐ Water

B Fill in the blanks.

1. Natural resources are the gifts of _____.
2. Sunlight, water and air are resources that can be used _____.
3. Petroleum taken out from the Earth is in a _____ form.
4. Wind energy can be used for producing _____.

C Write (T) for true and (F) for false sentences.

1. Natural resources are man-made things.
2. All natural resources have to be processed before they can be used.
3. Renewable resources can be used again and again.
4. We must conserve our natural resources.

☐
☐
☐
☐

D Answer the following questions.

1. What are natural resources? How are they important for all living beings?
2. What are renewable resources? Give examples.
3. What are non-renewable resources? Give examples.
4. How do human activities affect natural resources?

E Give reasons.

1. Some natural resources cannot be used in their natural form.
2. It is important to conserve natural resources.

Activities

Fun with Learning

(Arts Integration, Creativity)

Windmill Model

Use best out of the waste to make a working model of a simple windmill. Show it in the class.

